CBIR Skin Anomaly Retrieval

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1. Motivation

Content-Based Image Retrieval (CBIR) systems aim to retrieve images from a database that are visually similar to a query image. Instead of relying on textual annotations or metadata, CBIR analyses the visual content of images such as colour, texture, and structure. This approach is especially useful when textual descriptions are unavailable or insufficient.

A promising real-world application of CBIR is medical image retrieval for dermatology, specifically for skin anomaly detection and diagnostic support. Skin diseases often present subtle visual patterns. Dermatologists commonly compare new lesions with previously observed cases to assist diagnosis. However, manually searching through large image datasets is time-consuming and inefficient. A CBIR system can support clinicians by automatically retrieving visually similar skin lesion images from a database when a new image is provided as a query. The system can help in:

- assisting dermatologists in identifying similar cases
- supporting early detection of malignant lesions
- providing educational tools for medical training
- enabling large-scale dermatological image databases to be searched efficiently

In such a system, an input image of a skin lesion is analysed to extract visual descriptors. These descriptors are then compared with those of images stored in a database. The system returns the most visually similar images ranked by similarity score.

2. State of the Art

Recent research has shown that CBIR can be effectively applied to medical image analysis in dermatology. Esteva et al. [1] demonstrated that Convolutional neural networks (CNN) models trained on large dermatology datasets can achieve diagnostic performance comparable to dermatologists in skin cancer classification. Several studies have investigated CBIR systems specifically designed for dermoscopic image retrieval. Tschandl et al. [2] developed a CBIR approach based on neural network features that retrieves visually similar dermoscopic images from a database. CBIR provides interpretable results because clinicians can directly inspect visually similar examples rather than relying solely on probability outputs. Barhoumi and Khelifa [3] combined deep-learned feature extraction with transfer learning techniques and introduced a dynamic similarity metric selection mechanism. Instead of using a single fixed distance metric, their system predicts the most appropriate similarity measure for each query image, improving retrieval performance. Sadeghi et al. [4] developed an interactive dermoscopic CBIR system and evaluated its usability for supporting image interpretation and education. Their pilot study

demonstrated that users achieved higher diagnostic accuracy and confidence when visually similar images were presented alongside a query image.

3. CBIR adaptation to the domain of skin anomaly images

The CBIR implementation for skin anomaly images preserves the overall CBIR pipeline but adapts several components to better suit the domain of skin images. The original toy script used HOG and edge-direction histograms to represent structural shape and contour orientation. These descriptors are effective for generic object images where shape and edge layout are important features. However, skin anomaly images often contain more subtle visual cues, such as:

- pigmentation irregularities
- small texture variations
- internal lesion patterns
- heterogeneous colour distributions

In dermatological imagery, these local texture patterns are often more discriminative than global object structure. For this reason, HOG and the edge-direction histogram were replaced with a Local Binary Pattern (LBP) texture descriptor. LBP analyses the local neighbourhood of each pixel and encodes texture patterns based on intensity comparisons with neighbouring pixels (Figure 1). A histogram of these patterns represents the overall texture of the image.

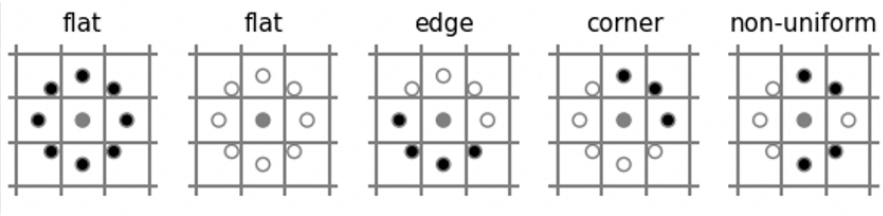

Figure 1: Local Binary Pattern (LBP) texture descriptor (see https://scikit-image.org/)

This makes LBP particularly suitable for skin anomaly retrieval because it captures micro-texture and surface irregularities more effectively than gradient-based descriptors such as HOG.

Furthermore, ResNet50 is replaced with EfficientNetB0 which provides a strong trade-off between performance and computational efficiency. Using EfficientNetB0 allows extraction of high-quality semantic embeddings while keeping computation relatively lightweight. These CNN features capture high-level visual patterns such as lesion appearance, shape patterns and global structural characteristics.

The final descriptor combines CNN semantic features, HSV color histograms and LBP texture descriptors. This combination better reflects the visual characteristics of dermatological images while preserving the structure of the original toy CBIR system.

4. Implementation of the CBIR System

The system follows the typical CBIR pipeline:

1. Image loading
2. Feature extraction
3. Feature fusion
4. Feature database construction
5. Indexing
6. Similarity search
7. Ranking and visualization

Each image in the dataset is converted into a numerical descriptor that represents its visual characteristics. During retrieval, the descriptor of a query image is compared to the descriptors stored in the database.

4.1 Feature Extraction

To represent images numerically, the system extracts several complementary feature types.

Neural Network Features The system uses a Convolutional Neural Network (CNN) to extract semantic image features. Specifically, the model EfficientNetB0 pretrained on ImageNet is used. The final classification layers are removed, and global average pooling is applied to obtain a compact feature vector representation. CNN features capture high-level visual information such as shape patterns, structural patterns and lesion appearance. These features represent the semantic content of the image.

Colour Histogram Descriptor Skin lesions often exhibit characteristic colour distributions. To capture this information, the system computes an HSV colour histogram. HSV colour space separates Hue (colour type), Saturation (colour intensity) and Value (brightness). The system computes histograms for each channel 32 bins for Hue, 32 bins for Saturation and 32 bins for Value. These histograms are concatenated into a 96-dimensional colour descriptor.

Texture Descriptor Skin lesions often differ in surface texture and micro-patterns. To capture these characteristics, the system uses Local Binary Patterns (LBP). LBP analyses the neighbourhood around each pixel and encodes the local texture pattern by comparing pixel intensities with neighbouring pixels. The resulting binary codes describe patterns such as edges, spots, and flat areas. A histogram of LBP codes is computed across the image, producing a compact texture representation. Texture descriptors are particularly useful in dermatology because they capture surface irregularities, lesion texture and micro-structures within the lesion.

4.2 Feature Fusion

The system combines the extracted descriptors using early fusion. Early fusion means that feature vectors are combined before similarity computation. The descriptors are concatenated into a single feature vector: **Final feature vector = CNN features | Color histogram | Texture descriptor.**

To balance the influence of each feature type, weighting factors are applied as controllable hyperparameters:

- CNN features: 0.60
- Colour histogram: 0.2
- Texture descriptor: 0.2

All vectors are normalized using L2 normalization before fusion to ensure that the different descriptors contribute proportionally to the final similarity computation.

4.3 Indexing and Similarity Search

To measure similarity between images, the system uses Euclidean distance (L2 distance) between feature vectors. Smaller distances indicate higher visual similarity. However, instead of comparing the query image with every image in the database (brute-force search), the system uses Non-Metric Space Library (NMSLIB) with HNSW indexing. HNSW (Hierarchical Navigable Small World) is a graph-based approximate nearest neighbour search algorithm. During indexing, feature vectors are organized into a graph structure where similar vectors are connected as neighbours. When a query image is processed:

1. The feature vector of the query image is extracted
2. The HNSW index is searched for the nearest feature vectors
3. Only a subset of the most promising candidates is explored
4. The nearest neighbours are returned

The HNSW indexing allows faster retrieval for larger datasets. Although approximate search may occasionally miss the exact nearest neighbour, in practice HNSW provides an excellent balance between speed and retrieval accuracy.

5. Result

The system returns the most similar images (Figure 2). The output includes query image, ranked similar images and similarity distances (Euclidean). Additionally, retrieved images are saved to a results folder together with a text file describing the ranking. This allows easy inspection of retrieval performance.

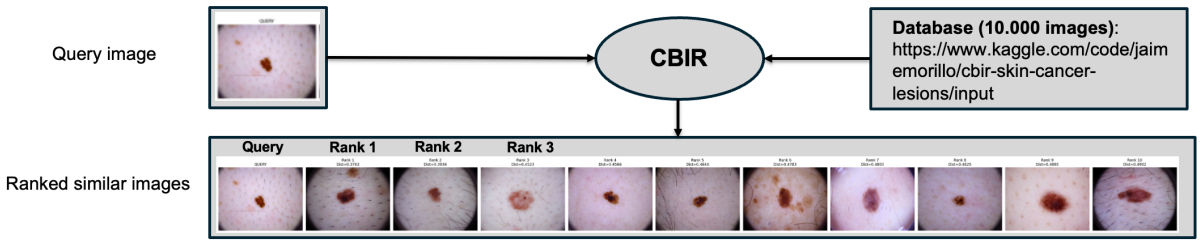

Figure 2: Most similar images

6. Conclusion

This project demonstrates how a CBIR system can be applied to the domain of dermatological image retrieval. The implemented system combines deep neural network features, colour histogram descriptors and texture descriptors using weighted early fusion and Euclidean distance similarity. Efficient similarity search is achieved through NMSLIB with HNSW indexing.

Compared with the original toy CBIR script, the system was adapted to better suit dermatological imagery by replacing gradient-based structural descriptors with LBP texture descriptors, which capture local lesion patterns more effectively. The approach enables retrieval of visually similar skin anomaly images and illustrates how CBIR techniques can support medical image analysis. Future improvements could include CNN models trained specifically on skin lesion images and evaluation using retrieval metrics such as precision and recall.

7. Data and notebook

https://github.com/SonkeKraft/CBIR-Skin-Anomalies.git

8. References

[1] Esteva, A., Kuprel, B., Novoa, R., et al. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542, 115–118, 2017.

[2] Tschandl, P., Argenziano, G., Razmara, M., & Yap, J. Diagnostic accuracy of content-based dermatoscopic image retrieval with deep classification features. *British Journal of Dermatology*, 181(1), 155–165, 2019.

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[4] Sadeghi, M., Chilana, P., Yap, J., Tschandl, P., & Atkins, M. S. Using content-based image retrieval of dermoscopic images for interpretation and education: A pilot study. *Skin Research and Technology*, 26(4), 503–512, 2020.